Solve a Graph using Depth First Search (DFS)

Aim :

To Implement DFS traversal on a graph to explore all reachable nodes from a source.

Procedure:

1. Define the graph using an adjacency list.
2. Start from the source node.
3. Recursively visit all unvisited neighbors.

Python Code:

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()

    visited.add(start)
    print(start, end=" ")

    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)


# Sample graph
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

print("DFS traversal from node A:")
dfs(graph, 'A')
```

Output:

DFS traversal from node A:
A B D E F C

Result:

Thus the program of implementing DFS traversal on a graph to explore all reachable nodes from a source. was executed successfully.